

DeskPRO Reports Manual

June 9th, 2014 • Revision #11

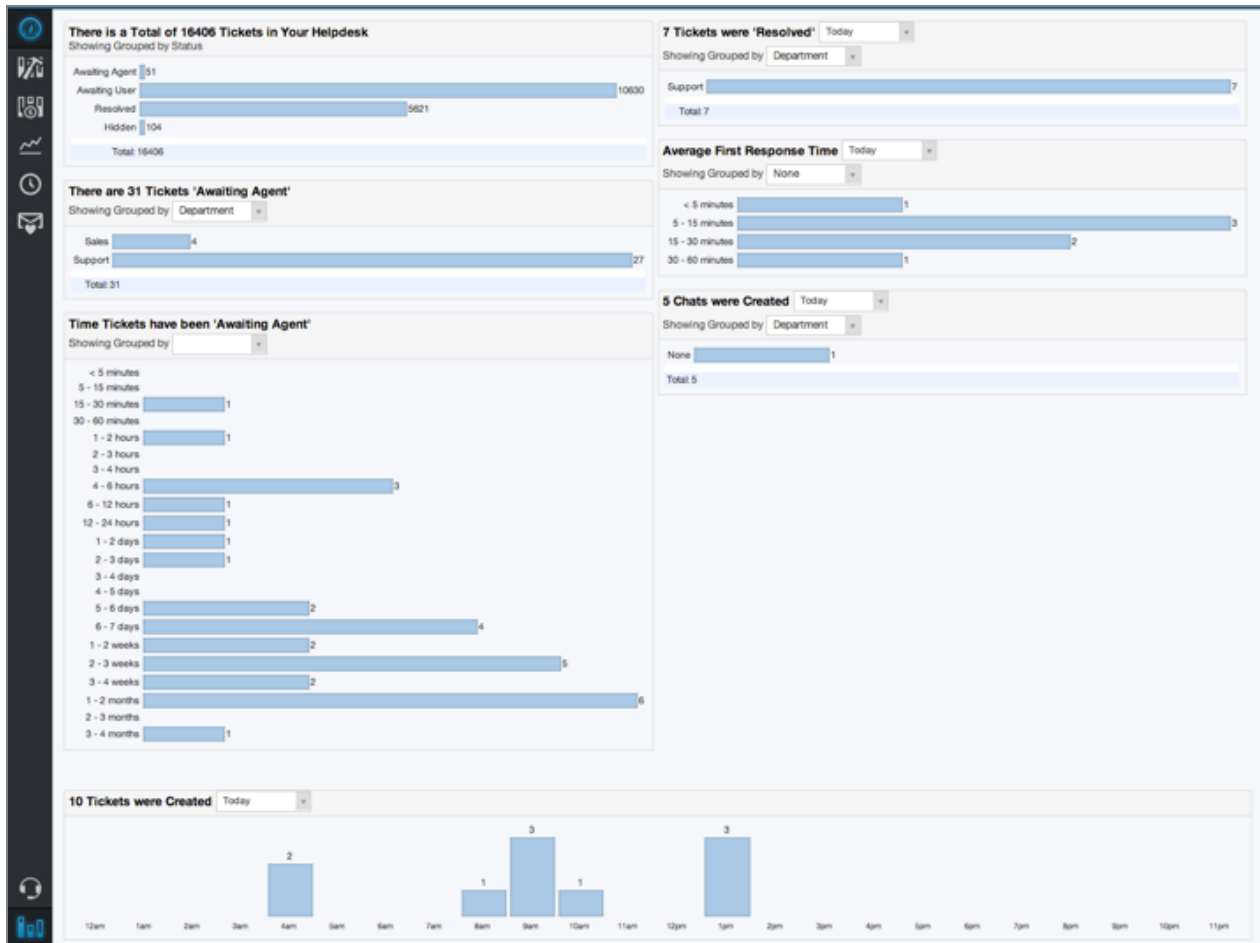
DeskPRO Reports Manual	1
1. DeskPRO Reports Manual	4
1.1. Using the Report Builder	5
1.2. Report Actions	8
2. DPQL Reference	9
2.1. General Expression Format	11
2.2. Date References and Comparisons	12
2.3. List of Placeholders	13
2.4. List of Functions	14
2.5. Accessing Custom Fields	17
2.6. Advanced: Dynamic Replacement	18
Date Ranges	18
Ticket Statuses	18
Field Groups	18
Ordering	19
3. APPENDIX: DPQL Field Reference	20
3.1. Tables	21
3.2. Data Types	23
Agent Team	23
Article	23
Article Attachment	24
Article Comment	24
Article Revision	25
Blob	25
Chat Conversation	26
Chat Message	27
Custom Data Article	27
Custom Data Feedback	28
Custom Data Organization	28
Custom Data Person	29
Custom Data Ticket	29
Custom Def Article	29
Custom Def Feedback	30
Custom Def Organization	31
Custom Def Person	31
Custom Def Ticket	32
Department	33
Download	33
Download Category	34

Download Comment	35
Download Revision	35
Email Gateway	36
Email Gateway Address	36
Email Source	36
Email Transport	37
Feedback	37
Feedback Attachment	38
Feedback Category	39
Feedback Comment	39
Feedback Revision	40
Feedback Status Category	40
Label Article	40
Label Blob	41
Label Chat Conversation	41
Label Download	41
Label Feedback	41
Label News	41
Label Organization	42
Label Person	42
Label Task	42
Label Ticket	42
Language	42
News	43
News Category	44
News Comment	44
News Revision	45
Organization	45
Organization Contact Data	46
Organization Email Domain	46
Person	46
Person Contact Data	48
Person Email	49
Person Email Validating	49
Person Pref	49
Person Usersource Assoc	50
Plugin	50
Product	50
Session	51
SLA	51
Task	52
Task Comment	53
Ticket	53
Ticket Access Code	56
Ticket Attachment	56
Ticket Category	57
Ticket Charge	57

Ticket Feedback	57
Ticket Log	58
Ticket Message	58
Ticket Participant	59
Ticket Priority	60
Ticket SLA	60
Ticket Trigger	60
Ticket Workflow	61
Usersource	61
Usersource Plugin	61
Visitor	62

1. DeskPRO Reports Manual

DeskPRO includes a powerful report system.



This manual explains how to use the report interface, focusing on the report builder which allows you to generate detailed custom reports.

1.1. Using the Report Builder

The report builder is powered by queries written in the DeskPRO Query Language (DPQL), which resembles SQL.

Any agent who can access the reports interface can access the report builder through this icon:

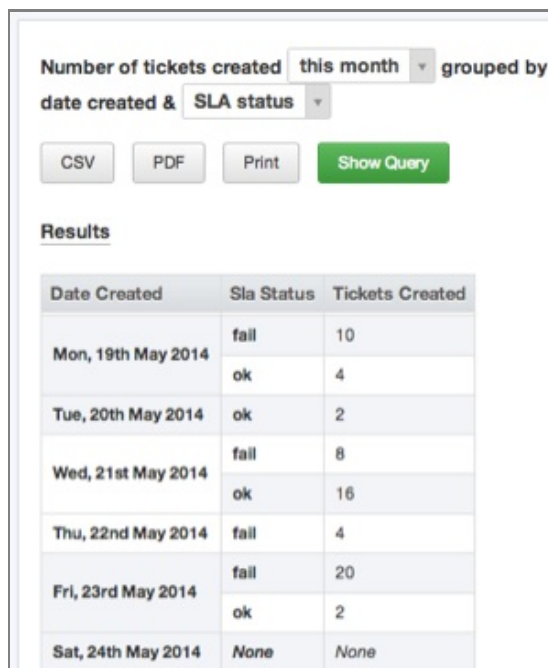


There are two types of reports in the report builder:

- **built-in:** specified by DeskPRO and cannot be edited directly
- **custom:** custom reports are specified by you or your agents

Reports can output different information in formats:

Tables - the most common output format is a table. In most cases, a simple table is generated, where a row represents one set of related data, such as the requested metrics for a particular department or agent.



Number of tickets created this month grouped by date created & SLA status

CSV PDF Print Show Query

Results

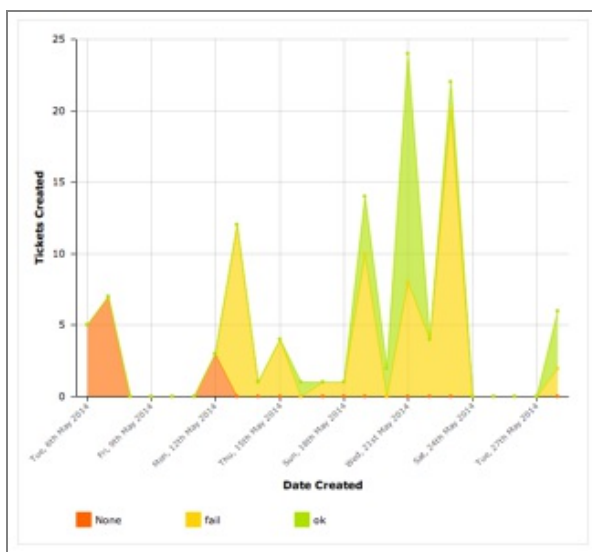
Date Created	Sla Status	Tickets Created
Mon, 19th May 2014	fail	10
	ok	4
Tue, 20th May 2014	ok	2
Wed, 21st May 2014	fail	8
	ok	16
Thu, 22nd May 2014	fail	4
Fri, 23rd May 2014	fail	20
	ok	2
Sat, 24th May 2014	None	None

Queries can also generate "matrix tables", where specific data is listed across the top and left side of the table (such as departments and agents) and the values in the table represent a particular metric for that combination (such as a specific agent in a specific department).

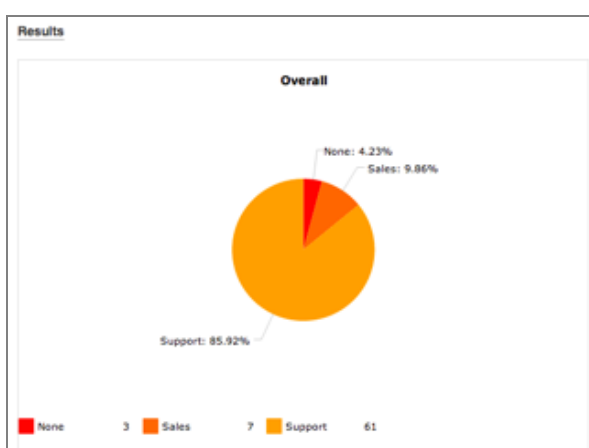
Results

Agent	Department			Total
	None	Sales	Support	
None	1	7	36	44
A-1 Sales Exec			1	1
A-1 Support Agent			2	2
A1 Admin	2		22	24
Total	3	7	61	71

Bar and line chart - some reports display charts, showing a visual relationship for the requested metrics across the possible values.



Pie chart - the final type of chart is a pie chart that can be used to visualize relationships as a percentage. All values for a metric will be displayed as slices of the pie with their size depending on their portion of the whole.



To view a report, simply click on the name of it in the left reports interface pane. It will be run immediately and the results will be displayed in the main pane.

Custom Reports

+ Create Custom Report

SLA failures by agent

Fire reports

Urgency pie

Tickets

Number of tickets created

this month

 grouped by

department

 &

agent

Number of tickets created

this month

 grouped by date created &

nothing

Number of tickets

awaiting agent

 grouped by

department

 &

agent

Number of tickets resolved

this month

 grouped by

department

 &

agent

Tickets awaiting agent split by

department

 ordered by

date created (ascending)

Tickets unresolved split by

department

 ordered by

date created (ascending)

Similar reports will be automatically grouped together. To choose the particular report you wish to view in a group, click the underlined portion and choose the type you want to run. Additionally, built-in reports are grouped into categories.

1.2. Report Actions

Once you are viewing a particular report, you can take a number of actions:

- **CSV** - clicking this button will output the data in the report to a Comma Separated Values file which can be imported into spreadsheet software like Excel. Please note that CSV files can only contain text, so charts will not be output.
- **Print** - sends the report to your printer with simplified formatting
- **PDF** - save the report as a PDF
- **Save to Clone** - copy this report, make changes, and save it to a new report. As the built-in reports cannot be edited directly, this allows you to create a customized version of a report.

Note that the PDF version of reports currently only include tables; charts are not shown in the PDFs.

2. DPQL Reference

Reports in the report builder are written using the DeskPRO Query Language, known as DPQL, which has a syntax that resembles SQL. Queries take this general form:

```
DISPLAY display types
SELECT select fields
FROM table
[WHERE conditions]
[SPLIT BY split fields]
[GROUP BY group fields]
[ORDER BY order fields]
[LIMIT amount [OFFSET offset amount]]
```

Clauses in square brackets are optional and italic text is a place holder for some other value.

DISPLAY

The display type can be a comma-separated value with one or two of the following values:

- TABLE - displays the results as a table
- BAR - displays the results as a bar chart
- LINE - displays the results as a line chart
- PIE - displays the results as a pie graph

For example, `DISPLAY TABLE` and `DISPLAY BAR, TABLE` are both valid.

SELECT

The select fields are a comma separated list of expressions that will be selected and displayed in the results table or graph.

In simple table output, the header for a column will automatically be derived from the DPQL to select that column. In some circumstances, this is not ideal, so aliases may be used to explicitly name the column. An alias takes the form of `expression AS 'alias'`, such as `COUNT() AS 'Total Tickets'`.

FROM

The FROM clause takes only a simple table name. See the [DPQL Field Reference](#) section for valid tables.

WHERE

The conditions in the WHERE clause are used to limit what data is displayed or used in calculations. Conditions can be joined by operators such as AND, OR, NOT, and parentheses to make complex expressions.

SPLIT BY

When split fields are provided, the results will be split into separate tables and graphs for each unique split field value. For example, you could list the open tickets for each agent in separate tables using

```
SPLIT BY tickets.agent
```

Split fields are a comma-separated list of expressions.

GROUP BY

The GROUP BY clause is used to group records that have the same values for the specified group fields, allowing you to do calculations such as determining totals and averages across those rows. For example, you could use this to return the total number of messages in particular tickets.

The group fields will automatically be displayed in the resulting table, so there is no need to add them to the SELECT clause.

Like in the SELECT clause, group fields can be aliased to change the name of the table header row.

To create a matrix table, the `X()` and `Y()` functions must be used. Each argument to `X()` controls the list of values going across the top of the table, while each `Y()` argument controls the values going down the left side.

ORDER BY

The ORDER BY clause determines how the rows will be ordered when they are returned. If no order is given, the results will be displayed in an undefined order.

The order fields are comma-separated expressions. Ordering will happen across the fields from left to right (ordering by the first expression, then using the second to resolve ties, and so on). Each expression may optionally have `ASC` or `DESC` appended to it to control whether ordering is in ascending or descending order. If no direction is specified, the expression will be ordered in ascending order.

The ORDER BY clause can access aliases that were specified in the SELECT or GROUP BY clauses using the syntax `@'alias'` (for example: `@'Total Tickets'`). Referencing an alias causes ordering to happen as if you wrote the actual expression in the ORDER BY clause.

LIMIT / OFFSET

The LIMIT clause allows you to limit the number of rows returned. If no limit *amount* is specified, a default limit of 2500 will be included to ensure correct operation.

The offset clause allows you to skip over a certain number of rows before returning the LIMIT *amount* of rows. The OFFSET defaults to 0.

2.1. General Expression Format

Most clauses of a DPQL statement accept a general expression format. Complex expressions are made up of smaller, simpler expressions. Expressions are made up of the following types of components:

- **Numbers** Numbers are simple, literal references to integer or decimal values.
- **Strings**
`'string' or "string"`
Strings are literal references to text. These will commonly be used in tests/comparisons and aliases.
- **Null**
`NULL` A null value will normally be used in tests where you want to check whether a value exists.
- **Parentheses**
`(expression)`
Parentheses are used to enforce a particular evaluation order, as in math.
- **Column References**
`table.col[.col2...]`
Column references are the actual references to data within DeskPRO. They control whether you are retrieving the ID or the subject of a ticket, for example. `tickets.subject` and `tickets.agent.name` are both valid column references. The DQPL equivalent of SQL joins are handled by column references that contain multiple column parts (such as `tickets.person.organization.name` which would get the name of the organization the person that started the ticket belongs to).
Note that all column references must start with the table name specified in the FROM clause. See the [DPQL Field Reference](#) section for a full list of columns.
- **Function calls**
`function([argument[, argument...]])`
Functions manipulate their arguments or produce a certain type of output. A list of available functions is below.
- **Comparisons** `expression [=, !=, >=, >, <, <=] expression`
- **Placeholders** `%placeholder%`
Placeholders are dynamic elements of a DPQL query that are automatically updated to be correct when the query is run. They will usually be combined with a comparison to ensure some condition is true, such as `tickets.created_date = %PAST_7_DAYS%` to only get tickets created in the last 7 days from when the query was run. A full [list of placeholders](#) is below.
- **Logic - AND, OR** `expression [AND, OR] expression`

2.2. Date References and Comparisons

You can reference specific dates directly in your queries, rather than using the relative date placeholders below. Dates can be referenced in two formats:

YYYY-MM-DD - This is just referring to a date, such as 2012-10-31. This implicitly has a time of 00:00:00 of the specified day.

YYYY-MM-DD HH:MM:SS - This refers to a date and a specific time in a 24-hour format, such as 2012-10-31 23:35:52.

Dates and times used in queries automatically reflect the timezone of the person running them.

This is an example DPQL query to list the tickets created from October 1st to 15th, 2012:

```
DISPLAY TABLE
SELECT tickets.id
FROM tickets
WHERE tickets.date_created >= '2012-10-01' AND tickets.date_created <= '2012-10-15 23:59:59'
```

2.3. List of Placeholders

%EVER%	A range covering any date.
%LAST_MONTH%	A range covering the entirety of the previous month. For example, if today is any day in September 2012, this placeholder will match all of August 2012.
%LAST_WEEK%	A range covering the entirety of the previous week, based on a new week starting on a Monday. For example, if today is Wednesday the 12th, this will match Monday the 3rd to Sunday the 9th.
%LAST_YEAR%	A range covering the entirety of the previous year. For example, if today is any day in 2012, this placeholder will match all of 2011.
%PAST_24_HOURS%	A range covering from exactly 24 hours ago until when the report is run.
%PAST_7_DAYS%	A range covering the 7 full days prior to the day when the report is run.
%PAST_30_DAYS%	A range covering the 30 full days prior to the day when the report is run.
%THIS_MONTH%	A range covering the entirety of the current month, from 00:00 on the first day until 23:59 on the last day.
%THIS_WEEK%	A range covering the entirety of the current week, from 00:00 on the first day until 23:59 on the last day. Weeks are assumed to start on a Monday and end on a Sunday
%THIS_YEAR%	A range covering the entirety of the current year, from 00:00 on the first day until 23:59 on the last day.
%TODAY%	A range covering the entirety of the today.
%TOMORROW%	A range covering the entirety of the tomorrow.
%YESTERDAY%	A range covering the entirety of the yesterday.

2.4. List of Functions

COUNT([condition])	Counts the total number of rows matched by the query or current group. If a condition is provided, counts the number of rows in the query or current group that match the condition.
COUNT_DISTINCT(expression)	Counts the total number distinct values for the given expression within the rows matched by the query or current group.
CURDATE()	Gets the current date in the time zone of the person running the query.
CURTIME()	Gets the current time in the time zone of the person running the query.
DATE_OFFSET_GROUP(seconds)	Groups a date offset/difference in seconds into human-readable ranges (0-15 minutes, 15-30 minutes, etc).
DATE_OFFSET_GROUP(to date, from date)	Displays the difference between the two provided dates as human-readable ranges (0-15 minutes, 15-30 minutes, etc).
FORMAT(value, format)	formats the value into the specified format. Possible formats include boolean, number, numberraw, datetime, date, time, year, percent, string.
LINK(value, link[, params..])	Links to value using the URL provided in link. Placeholders in link are replaced by the additional params. Placeholder values should be represented in sprintf format.
MATRIX(group X, group Y)	If both provided groups are non-null creates a matrix table from them. Otherwise, it creates a standard grouped table.
NOW()	Gets the current date and time in the time zone of the person running the query.
PERCENT(condition[, decimals])	determines the percentage of total rows or rows within the current group that match the condition The results are displayed as a percentage with decimals controlling the precision. 2 decimals are shown by default.
PERCENT(sql, printed)	gets the sql but ensures that printed is used if the value is ever going to be displayed. This is mostly helpful in the GROUP BY clause where you need to group on one expression but display the results of another.
TO_UTC(date)	converts date to UTC from the current person's time zone.
UTC(expression)	ensures that all dates and times within this function are calculated using UTC. This can increase performance.

A large number of functions are also available that have the exact same behavior as their MySQL equivalents:

- ABS
- AVG
- CEILING

- CONCAT
- CONCAT_WS
- CHAR_LENGTH
- DATE
- DATE_FORMAT
- DATEDIFF
- DAYNAME
- DAYOFMONTH
- DAYOFWEEK
- DAYOFYEAR
- FIELD
- FIND_IN_SET
- FLOOR
- FROM_UNIXTIME
- GREATEST
- HOUR
- IF
- IFNULL
- ISNULL
- LAST_DAY
- LEAST
- LEFT
- LENGTH
- LOCATE
- LOWER
- LPAD
- LTRIM
- MAX
- MIN
- MINUTE
- MONTH
- MONTHNAME
- POW
- QUARTER
- RAND
- REPEAT

- REPLACE
- REVERSE
- RIGHT
- ROUND
- RPAD
- RTRIM
- SECOND
- SQRT
- STDDEV_POP
- STDDEV_SAMP
- STRCMP
- SUBSTRING
- SUBSTRING_INDEX
- SUM
- TIME
- TIMESTAMP
- TRIM
- TRUNCATE
- UNIX_TIMESTAMP
- UPPER
- UTC_DATE
- UTC_TIME
- UTC_TIMESTAMP
- VAR_POP
- VAR_SAMP
- WEEKDAY
- WEEKOFYEAR
- YEAR

2.5. Accessing Custom Fields

Several content types in DeskPRO, including tickets and articles, support custom fields. The values set in these fields can be accessed via custom reports with a special syntax.

Each custom field is assigned an ID number. This will be listed with the custom fields in the admin interface. For example, for custom ticket fields, you can find this in **Tickets > Fields**. Make note of the ID as it will be needed in your query.

Once you have the ID, you can reference the value of a particular field via `custom_data[#]`. For example, if I were referring to the ticket custom field with ID 12, I'd use `tickets.custom_data[12]`.

This query will give a count of all tickets created this month per value for the custom field with ID 12:

```
DISPLAY TABLE
SELECT COUNT() AS 'Total Tickets'
FROM tickets
WHERE tickets.date_created = %THIS_MONTH%
GROUP BY tickets.custom_data[12]
```

Similarly, this will list the actual tickets created this month, split per value of the custom field with ID 12:

```
DISPLAY TABLE, BAR
SELECT tickets.id, tickets.subject, tickets.person
FROM tickets
WHERE tickets.date_created = %THIS_MONTH%
SPLIT BY tickets.custom_data[12]
```

2.6. Advanced: Dynamic Replacement

When creating custom reports, you can set up specific values to be dynamically replaced by a user selection. This is used by the built-in reports to allow users to choose a date range, grouping field, ticket status, or order.

Each dynamic replacement has two components:

- The title - this is how the user selects which value they want for the dynamic replacement. A default value should be specified here. The specific format varies for each replacement type, but the general format is `<#:replacement type[additional options]>` where # is a unique ID to identify the corresponding replacement in the query.
- The query - this is where the actual replacement is made to run the correct query. The general form is `%#:REPLACEMENT_TYPE[additional options]%` where # corresponds to the correct replacement in the title.

The available replacements are as follows:

Date Ranges

Title format	<code><#:date group, default: <i>default value</i>></code>
Possible defaults	today, yesterday, this_week, this-month, this_year, last_week, last_month, last_year, past_24_hours, past_7_days, past_30_days, ever
Partial title example	<code><1:date group, default: <i>this_month</i>></code>
Query format	<code>%#:DATE_GROUP%</code>
Partial query example	<code>tickets.date_created = %1:DATE_GROUP%</code>

Ticket Statuses

Title format	<code><#:status group:tickets, default: <i>default value</i>></code>
Possible defaults	awaiting_user, awaiting_agent, unresolved, resolved, hidden, any
Partial title example	<code><1:status group:tickets, default: awaiting_agent></code>
Query format	<code>%#:STATUS_GROUP:tickets:table_name%</code> (where <i>table_name</i> is how you access the tickets table in the query)
Partial query example	<code>%1:STATUS_GROUP:tickets:tickets_messages.ticket%</code>

Field Groups

Title format	<code><#:field group:tickets, default: <i>default value</i>></code>
--------------	---

Possible defaults	department, agent, agent_team, person, organization, language, useragency, category, priority, workflow, sla, sla_status, hour_created, day_week_created, day_month_created, month_created, year_created, <i>ticketfield#</i> (for custom ticket fields), <i>personfield#</i> (for custom person fields), <i>orgfield#</i> (for custom organization fields), none
Partial title example	grouped by <2:field group:tickets, default: department> & <3:field group:tickets, default: agent>
Query format	%#:FIELD_GROUP:tickets:table_name% (where <i>table_name</i> is how you access the tickets table in the query)
Partial query example	MATRIX(%2:FIELD_GROUP:tickets:tickets%, %3:FIELD_GROUP:tickets:tickets%)

Ordering

Title format	<#:field group:tickets, default: <i>default value</i> >
Possible defaults	date_created_asc, date_created_desc, last_agent_reply_asc, last_agent_reply_desc, last_user_reply_asc, last_user_reply_desc, total_waiting_asc, total_waiting_desc
Partial title example	ordered by <2:order group:tickets, default: date_created_asc>
Query format	%#:ORDER_GROUP:table_name% (where <i>table_name</i> is how you access the tickets table in the query)
Partial query example	%2:ORDER_GROUP:tickets%

3. APPENDIX: DPQL Field Reference

3.1. Tables

Table Name	Description
articles	Articles and information in the knowledgebase
article_attachments	Attachments to knowledgebase articles
article_comments	Comments for each knowledgebase article
chat_conversations	Records for each chat
chat_messages	Individual messages in each chat
downloads	Information about each file that has been specified as a download
download_comments	Individual comments for each download
feedback	Customer feedback and suggestion records
feedback_attachments	Attachments to feedback and suggestions
feedback_comments	Comments for feedback and suggestions
labels_articles	Labels for knowledgebase articles
labels_chat_conversations	Labels for chats
labels_downloads	Labels for downloads
labels_feedback	Labels for feedback and suggestions
labels_news	Labels for news entries
labels_organizations	Labels for organizations
labels_people	Labels for registered people
labels_tasks	Labels for tasks
labels_tickets	Labels for tickets
news	News entries
news_comments	Comments on news entries
organizations	Organizations
people	Registered people (users)
people_emails	Email addresses in use by registered people
tasks	Tasks
task_comments	Comments on tasks
tickets	Tickets
tickets_log	Ticket change log entries

tickets_messages	Individual messages in tickets
ticket_attachments	Attachments to tickets
ticket_charges	Ticket billing charges
ticket_feedback	Feedback on ticket responses
ticket_slas	SLA status records for tickets

3.2. Data Types

Agent Team

Field Name	Data Type
id	number
name	string

Article

Field Name	Data Type
content	string
date_created	datetime
date_end	datetime
date_published	datetime
end_action	string
hidden_status	string
id	number
language_id	number
num_comments	number
num_ratings	number
person_id	number
slug	string
status	string
title	string
total_rating	number
view_count	number
attachments	Article Attachment
custom_data	Custom Data Article
custom_data[#]	Custom Data Article (Gets data for the article field with ID #)
labels	Label Article
language	Language

person	Person
revisions	Article Revision

Article Attachment

Field Name	Data Type
article_id	number
blob_id	number
date_created	datetime
id	number
person_id	number
article	Article
blob	Blob
person	Person

Article Comment

Field Name	Data Type
article_id	number
content	string
date_created	datetime
email	string
id	number
ip_address	string
is_reviewed	boolean
name	string
person_id	number
status	string
validating	string
visitor_id	number
website	string
article	Article

person	Person
visitor	Visitor

Article Revision

Field Name	Data Type
article_id	number
content	string
date_created	datetime
id	number
person_id	number
status	string
title	string
article	Article
person	Person

Blob

Field Name	Data Type
authcode	string
blob_hash	string
content_type	string
date_cleanup	datetime
date_created	datetime
dim_h	number
dim_w	number
filename	string
filesize	number
id	number
is_media_upload	boolean
is_temp	boolean
original_blob_id	number

save_path	string
storage_loc	string
sys_name	string
title	string
labels	Label Blob
original_blob	Blob

Chat Conversation

Field Name	Data Type
agent_id	number
agent_team_id	number
date_assigned	datetime
date_created	datetime
date_ended	datetime
date_first_agent_message	datetime
department_id	number
ended_by	string
id	number
is_agent	boolean
is_window	boolean
person_email	string
person_id	number
person_name	string
rating_comment	string
rating_overall	number
rating_response_time	number
session_id	number
status	string
subject	string
total_to_ended	number

visitor_id	number
agent	Person
agent_team	Agent Team
department	Department
labels	Label Chat Conversation
messages	Chat Message
person	Person
session	Session
visitor	Visitor

Chat Message

Field Name	Data Type
author_id	number
content	string
conversation_id	number
date_created	datetime
id	number
is_html	boolean
is_sys	boolean
is_user_hidden	boolean
metadata	string
origin	string
person_name	string
author	Person
conversation	Chat Conversation

Custom Data Article

Field Name	Data Type
article_id	number
field_id	number

id	number
input	string
root_field_id	number
value	number
article	Article
field	Custom Def Article
root_field	Custom Def Article

Custom Data Feedback

Field Name	Data Type
feedback_id	number
field_id	number
id	number
input	string
root_field_id	number
value	number
feedback	Feedback
field	Custom Def Feedback
root_field	Custom Def Feedback

Custom Data Organization

Field Name	Data Type
field_id	number
id	number
input	string
organization_id	number
root_field_id	number
value	number
field	Custom Def Organization
organization	Organization

root_field	Custom Def Organization
------------	-------------------------

Custom Data Person

Field Name	Data Type
field_id	number
id	number
input	string
person_id	number
root_field_id	number
value	number
field	Custom Def Person
person	Person
root_field	Custom Def Person

Custom Data Ticket

Field Name	Data Type
field_id	number
id	number
input	string
root_field_id	number
ticket_id	number
value	number
field	Custom Def Ticket
root_field	Custom Def Ticket
ticket	Ticket

Custom Def Article

Field Name	Data Type
default_value	string
description	string

display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Article
parent	Custom Def Article
plugin	Plugin

Custom Def Feedback

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string

options	string
parent_id	number
plugin_id	number
sys_name	string
title	string
children	Custom Def Feedback
parent	Custom Def Feedback
plugin	Plugin

Custom Def Organization

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Organization
parent	Custom Def Organization
plugin	Plugin

Custom Def Person

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Person
parent	Custom Def Person
plugin	Plugin

Custom Def Ticket

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean

is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Ticket
parent	Custom Def Ticket
plugin	Plugin

Department

Field Name	Data Type
display_order	number
email_gateway_id	number
id	number
is_chat_enabled	boolean
is_tickets_enabled	boolean
parent_id	number
title	string
user_title	string
children	Department
email_gateway	Email Gateway
parent	Department

Download

Field Name	Data Type
blob_id	number
category_id	number
content	string

date_created	datetime
date_published	datetime
hidden_status	string
id	number
language_id	number
num_comments	number
num_downloads	number
num_ratings	number
person_id	number
slug	string
status	string
title	string
total_rating	number
view_count	number
blob	Blob
category	Download Category
labels	Label Download
language	Language
person	Person
revisions	Download Revision

Download Category

Field Name	Data Type
depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	Download Category

parent	Download Category
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Download Comment

Field Name	Data Type
content	string
date_created	datetime
download_id	number
email	string
id	number
ip_address	string
is_reviewed	boolean
name	string
person_id	number
status	string
validating	string
visitor_id	number
website	string
download	Download
person	Person
visitor	Visitor

Download Revision

Field Name	Data Type
blob_id	number
content	string
date_created	datetime
download_id	number
id	number
person_id	number
status	string

title	string
blob	Blob
download	Download
person	Person

Email Gateway

Field Name	Data Type
connection_options	string
connection_type	string
date_last_check	datetime
department_id	number
gateway_type	string
id	number
is_enabled	boolean
keep_read	boolean
linked_transport_id	number
title	string
addresses	Email Gateway Address
department	Department
linked_transport	Email Transport

Email Gateway Address

Field Name	Data Type
email_gateway_id	number
id	number
match_pattern	string
match_type	string
run_order	number
gateway	Email Gateway

Email Source

Field Name	Data Type
blob_id	number
date_created	datetime
error_code	string
gateway_id	number
headers	string
header_from	string
header_subject	string
header_to	string
id	number
object_id	number
object_type	string
source_info	string
status	string
uid	string
blob	Blob
gateway	Email Gateway

Email Transport

Field Name	Data Type
id	number
match_pattern	string
match_type	string
run_order	number
title	string
transport_options	string
transport_type	string

Feedback

Field Name	Data Type
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category_id	number
content	string
date_created	datetime
date_published	datetime
hidden_status	string
id	number
language_id	number
num_comments	number
num_ratings	number
person_id	number
popularity	number
slug	string
status	string
status_category_id	number
title	string
total_rating	number
validating	string
view_count	number
attachments	Feedback Attachment
category	Feedback Category
comments	Feedback Comment
custom_data	Custom Data Feedback
custom_data[#]	Custom Data Feedback (Gets data for the feedback field with ID #)
labels	Label Feedback
language	Language
person	Person
revisions	Feedback Revision
status_category	Feedback Status Category

Feedback Attachment

Field Name	Data Type
blob_id	number
date_created	datetime
feedback_id	number
id	number
person_id	number
blob	Blob
feedback	Feedback
person	Person

Feedback Category

Field Name	Data Type
depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	Feedback Category
parent	Feedback Category

Feedback Comment

Field Name	Data Type
content	string
date_created	datetime
email	string
feedback_id	number
id	number
ip_address	string
is_reviewed	boolean

name	string
person_id	number
status	string
validating	string
visitor_id	number
website	string
feedback	Feedback
person	Person
visitor	Visitor

Feedback Revision

Field Name	Data Type
content	string
date_created	datetime
feedback_id	number
id	number
person_id	number
status	string
title	string
feedback	Feedback
person	Person

Feedback Status Category

Field Name	Data Type
display_order	number
id	number
status_type	string
title	string

Label Article

Field Name	Data Type
article_id	number
label	string
article	Article

Label Blob

Field Name	Data Type
blob_id	number
label	string
blob	Blob

Label Chat Conversation

Field Name	Data Type
chat_id	number
label	string
chat	Chat Conversation

Label Download

Field Name	Data Type
download_id	number
label	string
download	Download

Label Feedback

Field Name	Data Type
feedback_id	number
label	string
feedback	Feedback

Label News

Field Name	Data Type
label	string
news_id	number
news	News

Label Organization

Field Name	Data Type
label	string
organization_id	number
organization	Organization

Label Person

Field Name	Data Type
label	string
person_id	number
person	Person

Label Task

Field Name	Data Type
label	string
task_id	number
task	Task

Label Ticket

Field Name	Data Type
label	string
ticket_id	number
ticket	Ticket

Language

Field Name	Data Type
base_filepath	string
flag_image	string
has_admin	boolean
has_agent	boolean
has_user	boolean
id	number
is_rtl	boolean
lang_code	string
locale	string
sys_name	string
title	string

News

Field Name	Data Type
category_id	number
content	string
date_created	datetime
date_published	datetime
hidden_status	string
id	number
language_id	number
num_comments	number
num_ratings	number
person_id	number
slug	string
status	string
title	string
total_rating	number
view_count	number

category	News Category
labels	Label News
language	Language
person	Person
revisions	News Revision

News Category

Field Name	Data Type
depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	News Category
parent	News Category

News Comment

Field Name	Data Type
content	string
date_created	datetime
email	string
id	number
ip_address	string
is_reviewed	boolean
name	string
news_id	number
person_id	number
status	string
validating	string

visitor_id	number
website	string
news	News
person	Person
visitor	Visitor

News Revision

Field Name	Data Type
content	string
date_created	datetime
id	number
news_id	number
person_id	number
status	string
title	string
news	News
person	Person

Organization

Field Name	Data Type
date_created	datetime
id	number
importance	number
name	string
picture_blob_id	number
summary	string
contact_data	Organization Contact Data
custom_data	Custom Data Organization
custom_data[#]	Custom Data Organization (Gets data for the organization field with ID #)
email_domains	Organization Email Domain

labels	Label Organization
picture_blob	Blob

Organization Contact Data

Field Name	Data Type
comment	string
contact_type	string
field_1	string
field_2	string
field_3	string
field_4	string
field_5	string
field_6	string
field_7	string
field_8	string
field_9	string
field_10	string
id	number
organization_id	number
organization	Organization

Organization Email Domain

Field Name	Data Type
domain	string
organization_id	number
organization	Organization

Person

Field Name	Data Type
can_admin	boolean

can_agent	boolean
can_billing	boolean
can_reports	boolean
creation_system	string
date_created	datetime
date_last_login	datetime
date_picture_check	datetime
disable_autoresponses	boolean
first_name	string
gravatar_url	string
id	number
importance	number
is_agent	boolean
is_agent_confirmed	boolean
is_confirmed	boolean
is_contact	boolean
is_deleted	boolean
is_disabled	boolean
is_user	boolean
is_vacation_mode	boolean
language_id	number
last_name	string
name	string
organization_id	number
organization_manager	boolean
organization_position	string
override_display_name	string
picture_blob_id	number
primary_email_id	number
summary	string

timezone	string
title_prefix	string
was_agent	boolean
contact_data	Person Contact Data
custom_data	Custom Data Person
custom_data[#]	Custom Data Person (Gets data for the person field with ID #)
emails	Person Email
labels	Label Person
language	Language
organization	Organization
picture_blob	Blob
preferences	Person Pref
primary_email	Person Email
usersource_assoc	Person Usersource Assoc

Person Contact Data

Field Name	Data Type
comment	string
contact_type	string
field_1	string
field_2	string
field_3	string
field_4	string
field_5	string
field_6	string
field_7	string
field_8	string
field_9	string
field_10	string
id	number

person_id	number
person	Person

Person Email

Field Name	Data Type
comment	string
date_created	datetime
date_validated	datetime
email	string
email_domain	string
id	number
is_validated	boolean
person_id	number
person	Person

Person Email Validating

Field Name	Data Type
auth	string
date_created	datetime
email	string
id	number
person_id	number
validating_content	string
person	Person

Person Pref

Field Name	Data Type
date_expire	datetime
name	string
person_id	number

value_array	string
value_str	string
person	Person

Person Usersource Assoc

Field Name	Data Type
data	string
date_created	datetime
id	number
identity	string
identity_friendly	string
person_id	number
usersource_id	number
person	Person
usersource	Usersource

Plugin

Field Name	Data Type
date_created	datetime
description	string
enabled	boolean
id	string
package_class	string
package_class_file	string
resources_path	string
title	string
version	string

Product

Field Name	Data Type
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depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	Product
parent	Product

Session

Field Name	Data Type
active_status	string
auth	string
data	string
date_created	datetime
date_last	datetime
id	number
is_bot	boolean
is_chat_available	boolean
is_helpdesk	boolean
is_person	boolean
page_count	number
person_id	number
visitor_id	number
person	Person
visitor	Visitor

SLA

Field Name	Data Type
active_time	string

allow_agent_manual	boolean
apply_all	boolean
apply_priority_id	number
apply_trigger_id	number
fail_trigger_id	number
id	number
sla_type	string
title	string
warning_trigger_id	number
work_days	string
work_end	number
work_holidays	string
work_start	number
work_timezone	string
apply_priority	Ticket Priority
apply_trigger	Ticket Trigger
fail_trigger	Ticket Trigger
ticket_slas	Ticket Sla
ticket_slas[#]	Ticket Sla (Gets ticket SLA data for the SLA with ID #)
warning_trigger	Ticket Trigger

Task

Field Name	Data Type
assigned_agent_id	number
assigned_agent_team_id	number
date_completed	datetime
date_created	datetime
date_due	date
id	number
is_completed	boolean

person_id	number
title	string
visibility	number
assigned_agent	Person
assigned_agent_team	Agent Team
comments	Task Comment
labels	Label Task
person	Person

Task Comment

Field Name	Data Type
content	string
date_created	datetime
id	number
person_id	number
task_id	number
person	Person
task	Task

Ticket

Field Name	Data Type
agent_id	number
agent_team_id	number
auth	string
category_id	number
creation_system	string
creation_system_option	string
date_agent_waiting	datetime
date_closed	datetime
date_created	datetime

date_feedback_rating	datetime
date_first_agent_assign	datetime
date_first_agent_reply	datetime
date_last_agent_reply	datetime
date_last_user_reply	datetime
date_locked	datetime
date_resolved	datetime
date_status	datetime
date_user_waiting	datetime
department_id	number
email_gateway_address_id	number
email_gateway_id	number
feedback_rating	number
has_attachments	boolean
hidden_status	string
id	number
is_hold	boolean
language_id	number
linked_chat_id	number
locked_by_agent	number
notify_email	string
notify_email_agent	string
notify_email_name	string
notify_email_name_agent	string
organization_id	number
person_email_id	number
person_email_validating_id	number
person_id	number
priority_id	number
product_id	number

ref	string
sent_to_address	string
status	string
subject	string
ticket_hash	string
total_to_first_reply	number
total_user_waiting	number
urgency	number
validating	string
waiting_times	string
workflow_id	number
worst_sla_status	string
access_codes	Ticket Access Code
agent	Person
agent_team	Agent Team
attachments	Ticket Attachment
category	Ticket Category
charges	Ticket Charge
custom_data	Custom Data Ticket
custom_data[#]	Custom Data Ticket (Gets data for the ticket field with ID #)
department	Department
email_gateway	Email Gateway
email_gateway_address	Email Gateway Address
labels	Label Ticket
language	Language
linked_chat	Chat Conversation
locked_by_agent	Person
messages	Ticket Message
organization	Organization
participants	Ticket Participant

person	Person
person_email	Person Email
person_email_validating	Person Email Validating
priority	Ticket Priority
product	Product
ticket_slas	Ticket Sla
ticket_slas[#]	Ticket Sla (Gets ticket SLA data for the SLA with ID #)
workflow	Ticket Workflow

Ticket Access Code

Field Name	Data Type
auth	string
id	number
person_id	number
ticket_id	number
person	Person
ticket	Ticket

Ticket Attachment

Field Name	Data Type
blob_id	number
id	number
is_agent_note	boolean
is_inline	boolean
message_id	number
person_id	number
ticket_id	number
blob	Blob
message	Ticket Message
person	Person

ticket	Ticket
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Ticket Category

Field Name	Data Type
display_order	number
id	number
parent_id	number
title	string
children	Ticket Category
parent	Ticket Category

Ticket Charge

Field Name	Data Type
agent_id	number
amount	number
charge_time	number
comment	string
date_created	datetime
id	number
organization_id	number
person_id	number
ticket_id	number
agent	Person
organization	Organization
person	Person
ticket	Ticket

Ticket Feedback

Field Name	Data Type
date_created	datetime

id	number
message	string
message_id	number
person_id	number
rating	number
ticket_id	number
person	Person
ticket	Ticket
ticket_message	Ticket Message

Ticket Log

Field Name	Data Type
action_type	string
date_created	datetime
details	string
id	number
id_after*	number
id_before	number
id_object	number
person_id	number
sla_id	number
sla_status	string
ticket_id	number
trigger_id	number
person	Person
sla	Sla
ticket	Ticket

- id_after means the ticket ID after it has changed; e.g. if this log represents a change from ID X to ID Y, id_after = Y.

Ticket Message

Field Name	Data Type
creation_system	string
date_created	datetime
email	string
email_source_id	number
id	number
ip_address	string
is_agent_note	boolean
message	string
message_full	string
message_hash	string
message_raw	string
person_id	number
show_full_hint	boolean
ticket_id	number
visitor_id	number
attachments	Ticket Attachment
email_source	Email Source
person	Person
ticket	Ticket
visitor	Visitor

Ticket Participant

Field Name	Data Type
access_code_id	number
default_on	boolean
id	number
person_email_id	number
person_id	number
ticket_id	number

access_code	Ticket Access Code
person	Person
person_email	Person Email
ticket	Ticket

Ticket Priority

Field Name	Data Type
id	number
priority	number
title	string

Ticket SLA

Field Name	Data Type
fail_date	datetime
id	number
is_completed	boolean
sla_id	number
sla_status	string
ticket_id	number
warn_date	datetime
sla	Sla
ticket	Ticket

Ticket Trigger

Field Name	Data Type
actions	string
date_created	datetime
event_trigger	string
event_trigger_options	string
id	number

is_enabled	boolean
run_order	number
sys_name	string
terms	string
terms_any	string
title	string

Ticket Workflow

Field Name	Data Type
display_order	number
id	number
title	string

Usersource

Field Name	Data Type
display_order	number
id	number
is_enabled	boolean
lost_password_url	string
options	string
source_type	string
title	string
usersource_plugin_id	number
usersource_plugin	Usersource Plugin

Usersource Plugin

Field Name	Data Type
adapter_class	string
form_model_class	string
form_template	string

form_type_class	string
id	number
plugin_id	number
title	string
unique_key	string
plugin	Plugin

Visitor

Field Name	Data Type
auth	string
date_created	datetime
date_last	datetime
email	string
id	number
ip_address	string
landing_page	string
last_page	string
name	string
person_id	number
ref_page	string
session_landing_page	string
user_agent	string
person	Person